# $F^2$AT: Feature-Focusing Adversarial Training via Disentanglement of Natural and Perturbed Patterns

Yaguan Qian, Chenyu Zhao, Zhaoquan Gu, Bin Wang, Shouling Ji, Wei Wang, and Yanchun Zhang

***Abstract*—Deep neural networks (DNNs) are vulnerable to adversarial examples crafted by well-designed perturbations. This could lead to disastrous results on critical applications such as self-driving cars, surveillance security, and medical diagnosis. At present, adversarial training is one of the most effective defenses against adversarial examples. However, in traditional adversarial training, it is still difficult to achieve a good trade-off between clean accuracy and robustness since DNNs still learn spurious features. The intrinsic reason is that traditional adversarial training makes it difficult to fully learn core features from adversarial examples when noise and examples cannot be disentangled. In this paper, we disentangle the adversarial examples into natural and perturbed patterns by bit-plane slicing. We assume the higher bit-planes represent natural patterns and the lower bit-planes represent perturbed patterns, respectively. We propose Feature-Focusing Adversarial Training ($F^2$AT), which differs from previous work in that it enforces the model to focus on the core features from natural patterns and reduce the impact of spurious features from perturbed patterns. The experimental results demonstrated that the clean accuracy and adversarial robustness with our $F^2$AT can be significantly improved.**

***Index Terms*—Adversarial example, adversarial training, robustness, trade-off.**

Received 21 December 2023; revised 20 May 2025; accepted 9 June 2025. Date of publication 16 June 2025; date of current version 24 July 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2024YFB3108100, and in part by the National Natural Science Foundation of China under Grant 62476250, Grant 62472335, Grant 62372137, Grant U21A20463, Grant U244120033, and Grant U24A20336. Recommended for acceptance by H. Hu. *(Corresponding authors: Zhaoquan Gu; Bin Wang.)*

Yaguan Qian is with the School of Artificial Intelligence and Information Engineering, Zhejiang University of Science and Technology, Hangzhou 310023, China (e-mail: qianyaguan@zust.edu.cn).

Chenyu Zhao is with the School of Science, Zhejiang University of Science and Technology, Hangzhou 310023, China (e-mail: 212209701014@zust.edu.cn).

Zhaoquan Gu is with the School of Computer Science and Technology, Harbin Institute of Technology (Shenzhen), Shenzhen 518071, China (e-mail: guzhaoquan@hit.edu.cn).

Bin Wang is with the Zhejiang Key Laboratory of Multidimensional Perception Technology, Application and Cybersecurity, Hangzhou 310052, China (e-mail: wbin2006@gmail.com).

Shouling Ji is with the School of Computer Science and Technology, Zhejiang University, Hangzhou 310058, China (e-mail: sji@zju.edu.cn).

Wei Wang is with the Ministry of Education Key Lab for Intelligent Networks and Network Security, Xi'an Jiaotong University, Xi'an 710000, China (e-mail: wei.wang@xjtu.edu.cn).

Yanchun Zhang is with the School of Computer Science and Mathematics, Victoria University, Melbourne, VIC 8001, Australia, and also with the School of Computer Science and Technology, Zhejiang Normal University, 321004, China (e-mail: yanchun.zhang@vu.edu.au).

Digital Object Identifier 10.1109/TKDE.2025.3580116

## I. Introduction

DESPITE the great success of deep neural networks (DNNs) in object recognition [1], machine translation [2], and speech translation [3], they are susceptible to adversarial examples added by carefully designed imperceptible noises. This could lead to disastrous results on critical applications like self-driving cars, surveillance security, and medical diagnosis. To address this challenge, various defense methods have been proposed, such as feature compression methods [4], defensive distillation [5], pruning [6], [7], and adversarial training [8], [9], [10], [11].

Adversarial training has been proved as one of the strongest empirical defenses against adversarial attacks [12]. The work in [13] showed that the models obtained from adversarial training have significantly higher feature concentration, which tends to reduce the weight of non-predictive and weakly predictive features. Unfortunately, the model after recent adversarial training may still learn spurious features [14], [15], [16], which prevents further improvements to the model's performance. One possible reason is that previous adversarial training was directly conducted on adversarial examples, without additional attention to the robust characterization of the adversarial examples. To further improve the generalization performance and robustness of the model, the model should learn more core features, i.e., robust features that contain real semantics, rather than spurious features.

To this end, Ilyas et al. [17] construct robust datasets that represent robust features. Their approach distinguishes these features based on whether their labels change following the addition of adversarial noise. However, this method still cannot avoid learning spurious features since the addition of noise to spurious features may also cause the label to change. For example, if the background is changed to a snowfield, a "Siberian Husky" may be misclassified as a "wolf" [18]. Tabacof et al. [19] pointed out that adversarial examples are not necessarily isolated, spurious points, and many adversarial images are located in dense regions of pixel space. Thus, our goal is to identify an approach that is efficient for learning core features and is easily integrated into adversarial training. This approach should not require an additional model for feature extraction as in [17], while also effectively preventing the acquisition of spurious features from dense regions. Interestingly, we found some extraordinary texture features occurring in the lower bit planes of

1041-4347 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 02:28:26 UTC from IEEE Xplore. Restrictions apply.

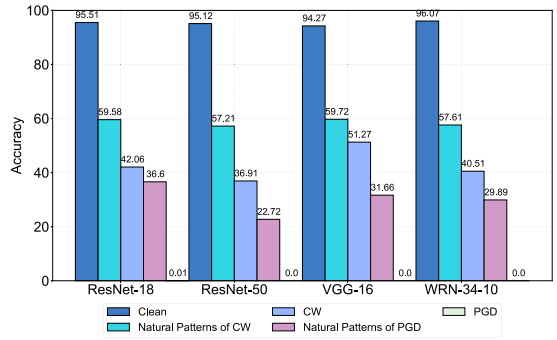



Fig. 1. Histogram of the accuracy of different models (clean training) on natural patterns. We find that the model well-trained on clean examples has very low robustness to PGD attacks, with its highest robustness accuracy of 0.01%, and slightly higher robustness to CW attacks, with a maximum of 51.27%. In contrast, the accuracy of these models against the natural patterns of the adversarial examples is substantially improved, most notably by ResNet-18.

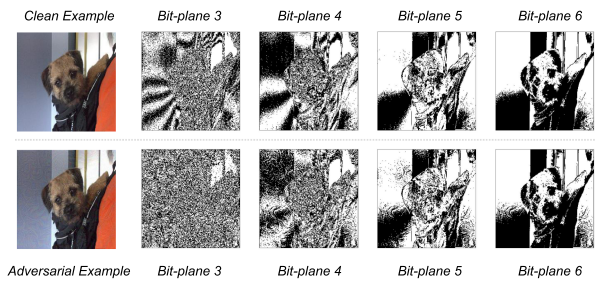



Fig. 2. Examples of bit-planes of the clean and adversarial images. The adversarial image is generated by the standard PGD20 attack with a perturbation budget of 8/255. The black image represents the bit-plane with all pixel values filled with zeros.

the adversarial example compared to clean examples, but this phenomenon is not obvious in the upper bit planes (see Fig. 2). This finding inspires us to disentangle an adversarial image into two patterns, natural and perturbed patterns, by bit-plane slicing which not rely on model's prediction. Specifically, upper bit-planes correspond to natural patterns, and lower bit-planes correspond to perturbed patterns. We hope that the model learns more core features in natural patterns and fewer spurious features in perturbed patterns to improve both the model's clean accuracy and adversarial robustness. We demonstrate the need to learn core features in natural patterns (see Fig. 1). As shown in Fig. 1, we evaluate several clean models with different architectures under adversarial attacks (e.g., PGD). Although these models show almost no robustness on PGD (average robust accuracy under 0.01%), their performance on the higher bit-planes (natural patterns) increases significantly, reaching over 30% on average. This indicates that the higher bit-planes tend to preserve semantically meaningful and robust features that are less sensitive to adversarial perturbations.

Although the robustness achieved solely through data preprocessing improves over 30% on average, it remains far from sufficient. It is still significantly lower than standard adversarial training (SAT), which typically achieves robust accuracy over 50%. Therefore, we further incorporate the disentanglement into the adversarial training process, enabling the model to learn core features from the natural pattern more effectively during training.

To this end, we proposed a novel adversarial training method, named Feature-Focusing Adversarial Training ($F^2$AT), based on natural and perturbed patterns, which is different from existing adversarial training [8], [9], [20], [21], [22], [23] in that they did not further disentangle adversarial examples to reduce the interference from adversarial and natural perturbations. Through this disentanglement, on the one hand, we enforce the model to focus more on the natural patterns and less on perturbed patterns; on the other hand, we want the decision boundaries to be adjusted depending more on natural patterns while less on adversarial examples, which can avoid over-adjustment of the decision boundaries [24]. Correspondingly, we designed two loss regularizers, i.e., pattern-dependent loss and natural-margin loss. (1) For *pattern-dependent loss*, we employ mutual information (MI) to measure the dependence between the extracted features and the natural (perturbed) pattern. Then, we amplify the MI between the natural patterns and features, while diminishing the MI between the perturbed patterns and features. (2) For *natural-margin loss*, it adopts the margin between the decision boundary and the natural patterns because these patterns are devoid of any perturbations.

$F^2$AT can improve both clean accuracy and adversarial robustness by learning more core features. The experiments prove that the method is effective in defending against most adversarial attacks.

Our contributions are summarized as follows:

- We disentangle an adversarial example into natural and perturbed patterns by bit-plane slicing, which provides the possibility for adversarial training to focus on core features to improve clean accuracy and robustness.
- We propose a new adversarial training method ($F^2$AT) based on this disentanglement. Two loss regularizers are developed to make the model focus on the core features of natural patterns.
- The extensive experiments under both white-box and black-box settings are conducted on CIFAR-10, CIFAR-100, and Tiny-ImageNet, which shows that clean accuracy and adversarial robustness with our adversarial training can be significantly improved.

## II. Related Work

### A. Adversarial Attacks

Szegedy et al. [25] first proposed that adversarial examples can mislead deep neural networks, lots of effective adversarial attack methods, such as Fast Gradient Sign Method (FGSM) [8], Projected Gradient Descent Attack (PGD) [9], and Carilini and Wagner Attack (CW) [26] are proposed. Croce et al. [27] chose several attack variants to constitute AutoAttack (AA). Existing attack methods can be divided into white-box attacks and black-box attacks. White-box attacks know all the information about the target model when generating adversarial examples, and black-box attacks know only part of the attacked model's output when generating adversarial examples. In general, black-box attacks simulate the model gradient by repeatedly querying the

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 02:28:26 UTC from IEEE Xplore. Restrictions apply.

target model (query-based attack) [28], [29] or searching for an alternative model similar to the target model (transfer-based attack) [30], [31], [32], [33]. Since attackers hardly know the parameters of the target model in practical applications, the model's performance against black-box attacks can better reflect the real robustness. In [34], [35], [36], [37], we have seen multiple models compromised by subsequent, more powerful attacks. This motivates the research community to evaluate defenses against these attacks, which are specifically designed to trick defense mechanisms.

### B. Adversarial Defense

Adversarial defense is divided into passive and active defense. Passive defense refers to the modification of model inputs to defend against adversarial attacks, e.g., denoising [38], and detection of adversarial examples [39]. Active defense refers to the modification of model parameters to defend against adversarial attacks, e.g., adversarial training. Madry et al. [40] formulate adversarial training as a minimax optimization problem. After that, much work has been proposed to further improve the robustness of adversarial training. For example, Zhang et al. [20] achieve this by minimizing the dissimilarity between original and adversarial examples. Kannan et al. [41] considered the logit results of the model produced by both clean and adversarial examples. They created a regularization term by reducing the distance between the logit pairs. Wang et al. [21] considered the situation when the train examples were correctly or incorrectly classified initially and designed two regularizers for each case. Mao et al. [42] used a triplet loss to push examples in different classes further apart and pull examples in the same class closer together. Zhong et al. [43] used a margin-based approach, considering the margin between clean and adversarial examples.

However, the adversarial examples in these adversarial training are typically generated from clean examples and share the same class labels. If both the labels of clean examples and adversarial examples are simultaneously utilized in the loss function as in the methods mentioned in literature [20] and [21], the model may learn to distinguish them based on the label information rather than genuinely capturing the underlying data characteristics [44]. This can lead to an improvement in the model's performance on adversarial examples but may result in reduced robustness in real-world applications. Some methods can be applied to solve these problems. For example, SEAT [45] does not invoke any regular terms, and robustness improvement can be achieved by smoothing the weights of historical models along the optimization trajectory. However, these methods make it difficult to avoid interference from spurious features of adversarial examples, making it difficult to learn core features.

Therefore, we propose a new adversarial training method that can make a model focus on the core features of images by disentanglement of natural and perturbed patterns. Our method can improve the classification accuracy and robustness of the model at the same time.

## III. Preliminary

In this paper, we use mutual information (MI) to measure the dependence between the extracted features and the natural (perturbed) pattern. MI can be thought of as the amount of information contained in one random variable about another random variable, or the reduction of uncertainty in one random variable due to another random variable. Before introducing the definition of mutual information, we first introduce the definition of entropy. Entropy is a measure of uncertainty of a random variable: $H(X) = \sum_x p(x) \log p(x)$ where $x$ is a specific value of $X$ and $p(x)$ is the marginal probability of $X$. The joint entropy of $X$ and $Y$, with joint mass probability $p(x, y)$, is the sum of the uncertainty contained by the two variables. Formally, joint entropy is defined as: $H(X, Y) = \sum_{x,y} p(x, y) \log p(x, y)$. The conditional entropy is defined as:

$$H(X|Y) = -\sum_{x,y} p(x, y) \log p(x|y). \tag{1}$$

The definition of MI relies on the idea of entropy and measures the level of information transfer and association between variables. It calculates the difference between the entropy of individual variables and the joint or conditional entropy. Let $X$ and $Y$ be random variables, the MI of $(X, Y)$ is defined as follows:

$$I(X; Y) = \sum_{x,y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}. \tag{2}$$

According to the definition, if $X$ and $Y$ are independent of each other, then $X$ provides no information about $Y$ and vice versa, $I(X; Y) = I(Y; X) = 0$. At the other extreme, if $X$ is bijective to $Y$, then all the information passed is shared by $X$ and $Y$. In this case, $I(X; Y) = H(X) = H(Y)$.

To better understand our proposed Theorem 2 in Section IV-C, we provide the following lemmas to explore the general relationship between MI and the information entropy:

*Lemma 1:* Let $X$ and $Y$ be random variables, we have

$$\begin{aligned} I(X; Y) &= H(X) - H(X|Y) = H(Y) - H(Y|X) \\ &= H(X) + H(Y) - H(X, Y), \end{aligned} \tag{3}$$

where $I(X; Y)$ denotes the MI of $X$ and $Y$, $H(X|Y)$ is the conditional entropy of $X$ and $Y$, $H(X, Y)$ is the joint entropy of $X$ and $Y$.

*Lemma 2:* Let $X$, $Y$, and $F$ be three random variables, we have

$$H(Y|X) = H(Y|X, F) + I(F; Y|X), \tag{4}$$

$$I(X; F) = I(X; F|Y) + I(X; Y; F). \tag{5}$$

Information entropy and mutual information provide a precise mathematical tool to prove the rationality of our method. The proof details of Lemmas 1 and 2 can be found in reference [46].

## IV. Proposed Method

### A. Motivation

Traditional adversarial training directly incorporates adversarial examples without disentangling them into perturbed and natural patterns. Consequently, these approaches inevitably lead to learning spurious features. In other words, the core features are obscured by adversarial perturbations, rendering it challenging for the model to focus on them. However, we found

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 02:28:26 UTC from IEEE Xplore. Restrictions apply.

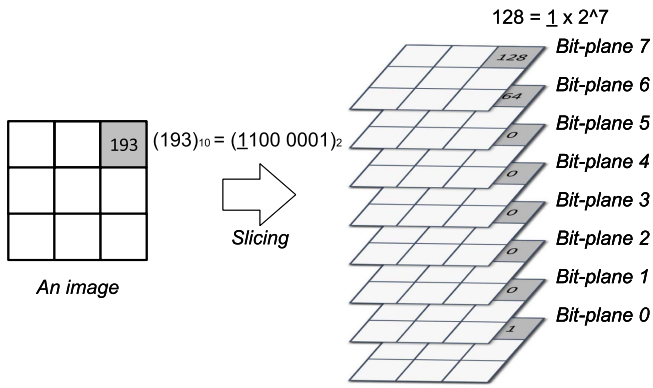



Fig. 3. An example of bit-plane slicing applied to an 8-bit single-channel image with the size of $3 \times 3$. 193 is the decimal value of a pixel in the upper-right corner of the image.

that adversarial perturbations have little impact on the pixels of higher bit-plane as shown in Fig. 2. This inspired us to explore the concept of bit-plane. By performing the bit-plane slicing operation, we can achieve the disentanglement of natural patterns and perturbed patterns. Considering that bit-plane and bit-plane slicing are two important concepts in this paper, we present their formal definitions as follows.

*Definition 1 (Bit-plane and Bit-plane Slicing):* For an $N$-bit single-channel image $X \in \mathbb{R}^{d \times d}$, each pixel value can be represented as a combination of $N$ independent bits in its binary form. The spatial maps corresponding to a specific bit position $i$ $(0 \leq i < N)$ are referred to as the image's bit-planes, denoted as $X^{(i)} \in \{0, 2^i\}^{d \times d}$. The bit-plane slicing operation decomposes the image $X$ into a set of bit-plane layers $\{X_{bp}^{(i)}\}_{i=0}^{N-1}$, where $X_{bp}^{(i)}$ represents the $i$th bit-plane of the image $X$. Note that $X_{bp}^{(i)} = 2^i \cdot (\lfloor \frac{X}{2^i} \rfloor \bmod 2)$, where $\lfloor \cdot \rfloor$ represents the floor function (rounding down to the nearest integer).

This decomposition ensures that the original image can be reconstructed by $X = \sum_{i=0}^{N-1} X_{bp}^{(i)}$. To help understand the bit-plane slicing, we provide an example as shown in Fig. 3. Assuming that a pixel in the image has the value of 193, its binary representation is "1100 0001". The corresponding values in bit-planes 7, 6, and 0 are $2^7$, $2^6$, and $2^0$, respectively, and the values in the rest bit-planes are 0. Therefore, the $i$th bit-plane is a collection of $2^i$ or 0.

Large-scale features make up the majority of how people perceive images, and adding finer details only serves to amplify this view [47], [48]. The human visual system inherently prioritizes information from higher bit-planes because of this underlying knowledge of giving more weight to information present at the higher bit-planes. This makes it naturally resistant to adversarial perturbations primarily affecting lower bit-planes. In contrast, deep neural networks lack this intrinsic bias, allowing adversarial perturbations to alter prediction categories.

Fig. 2 intuitively illustrates why bit-plane slicing is effective in revealing the knowledge concealed by adversarial perturbations. The lower bit-planes primarily contain more textural details and are more susceptible to adversarial noise [49]. Conversely, the higher bit-planes retain clearer structural information that is less prone to adversarial interference. For example, adversarial attacks typically operate within a small perturbation budget, such as 8/255. Given a pixel with an initial value of 193, its binary representation changes from "11000001" to "11001001" under perturbation. Notably, the values in the higher bit-planes (7th and 6th) remain unaffected, as the perturbation primarily impacts the lower bit-planes. These findings underscore the importance of leveraging bit-plane decomposition to enhance model robustness. We hope the DNN learns more *core* features at the higher bit-planes and fewer *spurious* features at the lower bit-planes. Therefore, we disentangle an image into two patterns, natural and perturbed patterns, by bit-plane slicing.

### B. Natural and Perturbed Patterns

Let $\mathcal{X}$ be a natural image space, and $\mathcal{Y}$ be a label space, existing a ground-truth labeling function: $g : \mathcal{X} \to \mathcal{Y}$. The classification model $h$ is to approximate the ground-truth labeling function $g$. Let $h(X; w, \psi) = T_w \circ E_\psi(X)$, where $X \in \mathcal{X}$ is an image, $w$ and $\psi$ are the model parameters. The function $E_\psi : \mathcal{X} \to \mathcal{F}$ represents a feature extractor where $\mathcal{F}$ denotes the feature space. The function $T_w : \mathcal{F} \to \mathcal{Y}$ represents a classifier. Suppose $X'$ is an adversarial example generated on $X$, i.e, for a clean image $X \in \mathcal{X}$, its adversarial example $X' = X + \delta$, such that $h(X') \neq y$ but $g(X') = y$, where $y \in \mathcal{Y}$ is the ground-truth label of $X$, and $\delta$ is the adversarial perturbation with constraint $\|\delta\|_p < \epsilon$. Here, $\|\delta\|_p$ is the $p$-norm and $\epsilon$ is a perturbation budget. Next, we formally define natural and perturbed patterns as follows:

*Definition 2 (Natural Patterns):* For an image $X$, its natural pattern $X_{nat}$ refers to the component of the pixels employed by a feature extractor $E_\psi$ in the unperturbed condition. The natural pattern not only correlates with the ground-truth label $y$ but also are relevant to a ground-truth labeling function $g$.

*Definition 3 (Perturbed Patterns):* The contaminated part of the pixels is referred to as a perturbed pattern of $X$, denoted by $X_{pert}$, which will interfere with a feature extractor $E_\psi$ and mislead a classifier to incorrect class labels.

According to the above definitions, it is difficult to disentangle an adversarial example into natural and perturbed patterns strictly. Recall that lower bit-planes are easily contaminated by adversarial perturbation in Section IV-A. Bit-plane slicing provides a feasible way to disentangle adversarial examples into natural and perturbed patterns to the greatest extent.

Suppose $X$ is an $R$-bit image, we set the hyperparameter $K$ to slice the image. We accumulate the $0 \sim (R - K - 1)$ bit-planes of the three RGB channels to get the *perturbed pattern* $X'_{pert}$ and add up the remaining upper bit-planes to obtain the *natural pattern* $X'_{nat}$. Note that for $X'_{pert}$, $K \sim (R - 1)$ bit-planes are filled with zero. Similarily, for $X'_{nat}$, $0 \sim (R - K - 1)$ bit-planes are filled with zero. Accordingly, we have $X' = X'_{nat} + X'_{pert}$.

### C. Feature Dependence

We expect the adversarial training to make the model focus more on the core features of natural patterns and less on the spurious features of perturbed patterns. In other words, we

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 02:28:26 UTC from IEEE Xplore. Restrictions apply.

hope to reduce the noise interference on extracted features from adversarial examples. For better understanding, we give a formal definition of core features:

*Definition 4 (Core Feature):* The core features refer to the features with stable semantics and strong class discrimination extracted from the natural pattern. Formally, for an image $X$, the natural pattern is formulated by bit-plane slicing as $X_{nat} = \sum_{i=K}^{R-1} X_{bp}^{(i)}$, and the corresponding core features are denoted as $E_\psi(X_{nat})$.

Since adversarial examples are re-generated in each training iteration, the natural patterns derived from them vary slightly. The learning objective is therefore to capture invariant and stable semantics across different adversarial variants of the same clean image. This consistency helps the model generalize better and resist overfitting to spurious adversarial artifacts. Unlike adversarial features, which may encode perturbed patterns specific to a particular perturbation, core features rely on the more stable higher bit-planes, which are less sensitive to perturbations. Thus, in our adversarial training framework, the model primarily learns core features from adversarial examples, that is, $E_\psi(X'_{nat})$.

Considering mutual information (MI) is a good measure of the dependence between two sets [50], we intend to adopt MI to explicitly measure the dependence between an adversarial image $X'$ and its feature $F'$. Recall our goal is to reduce the dependence on perturbed patterns in the adversarial training, we use $-I(X';F')$ as a regular term added to the loss function. However, we find simply minimizing $-I(X';F')$ in the loss function cannot achieve our expected goal. Since the loss does not guide the model to make a distinction between the learning of core features and noise. For this reason, we disentangle an adversarial image into perturbed and natural patterns by bit-plane slicing, i.e., $X' = X'_{nat} + X'_{pert}$, as described in Section IV-B. Similarly, we disentangle $I(X';F')$ into $I(X'_{nat};F')$ and $I(X'_{pert};F')$ for better guidance, where $I(X'_{nat};F')$ and $I(X'_{pert};F')$ represent the MI of the natural and perturbed patterns to the adversarial example feature maps, respectively. We will prove the feasibility of this disentanglement by the bit-plane slicing to approximate MI linearly. Zhou et al. [50] has given the transformation relation of MI in four variables as shown in Theorem 1.

*Theorem 1:* Let $X'$, $X'_{nat}$, $X'_{pert}$, $F'$ be four random variables, such that $X' = X'_{nat} + X'_{pert}$, we have

$$\begin{aligned} I(X';F') &= I(X'_{nat};F') + I(X'_{pert};F') - H(F'|X') \\ &\quad + H(F'|X'_{nat},X'_{pert}) - I(X'_{nat};X'_{pert};F'). \end{aligned} \tag{6}$$

Unfortunately, computing $H(F'|X')$, $H(F'|X'_{nat},X'_{pert})$, and $I(X'_{nat};X'_{pert};F')$ is non-trivial [51]. Therefore, we tend to construct a surrogate for $I(X';F')$. We find if $X' = X'_{nat} + X'_{pert}$ is implemented through bit-plane slicing, $I(X';F')$ can be lineally approximated by $I(X'_{nat};F')$ and $I(X'_{pert};F')$. Theorem 2 provides evidence to implement this idea as follows:

*Theorem 2:* Let $X'$ be an adversarial example, and $F'$ be the features of $X'$. If $X'_{nat}$ and $X'_{pert}$ are the natural and perturbed patterns of the adversarial examples $X'$ generated by bit-plane slicing, respectively, then we have

$$-H(F'|X') + H(F'|X'_{nat},X'_{pert}) - I(X'_{nat};X'_{pert};F') \approx 0. \tag{7}$$

*Proof 1:* To establish approximate equivalence, we first prove that the conditional entropy of the feature $F'$ given the natural and perturbed patterns is similar to the conditional entropy given only the adversarial examples. According to the definition of conditional entropy in (1), we have

$$\begin{aligned} &H(F'|X'_{nat},X'_{pert}) \\ &= -\sum_{X'\in\mathcal{D}'} p(F',X'_{nat},X'_{pert}) \log_2 p(F'|X'_{nat},X'_{pert}), \end{aligned} \tag{8}$$

where $\mathcal{D}'$ denotes the adversarial example set, $p(F',X'_{nat},X'_{pert})$ represents the joint probability, and $p(F'|X'_{nat},X'_{pert})$ is the conditional probability of $F'$ given $X'_{nat}$ and $X'_{pert}$. Similarly, we have

$$H(F'|X') = -\sum_{X'\in\mathcal{D}'} p(F',X') \log_2 p(F'|X'). \tag{9}$$

Notice that for each adversarial example, there is a unique counterpart of the natural and perturbed pattern, i.e., $X' = X'_{nat} + X'_{pert}$ is established. We can express the probability of $X'$ in terms of $X'_{nat}$ and $X'_{pert}$, that is, $p(X') = p(X'_{nat},X'_{pert})$. Substituting this relation into (9), we have

$$\begin{aligned} H(F'|X') &= -\sum_{X'\in\mathcal{D}'} p(F',X') \log_2 p(F'|X') \\ &= -\sum_{X'\in\mathcal{D}'} p(F',X'_{nat},X'_{pert}) \log_2 p(F'|X'). \end{aligned} \tag{10}$$

This is the same as (8). Therefore, we obtain

$$H(F'|X'_{nat},X'_{pert}) = H(F'|X'). \tag{11}$$

It means $-H(F'|X') + H(F'|X'_{nat},X'_{pert}) = 0$ in (7). Next, we should prove $I(X'_{nat};X'_{pert};F') \approx 0$ in (7).

Suppose $\xi$ and $c$ are constants such that $\xi \ll c$ and $|I(X'_{nat};X'_{pert}) - c| \le \xi$, and $I(X'_{nat};F') \neq 0$, $I(X'_{pert};F') \neq 0$, then for any given $\eta > 0$, there exists a $\eta' > 0$ such that for all $X'_{pert}$ satisfying $I(X'_{pert},F') < \eta'$, it implies $I(X'_{nat},F') > \eta$, where $\eta'$ is a constant as small as possible. Further applying Lemmas 1 and 2, we have

$$\begin{aligned} &I(X'_{nat};X'_{pert};F') \\ &= I(X'_{pert};F') - \big(H(X'_{pert}|X'_{nat}) - H(X'_{pert}|X'_{nat}F')\big) \\ &\le H(X'_{nat}|F') + I(X'_{pert};F') - H(X'_{pert}|X'_{nat}) \\ &= H(X'_{nat}) - I(X'_{nat}|F') + I(X'_{pert},F') - H(X'_{pert}|X'_{nat}). \end{aligned} \tag{12}$$

Taking $\eta = H(X'_{nat}) - H(X'_{pert}|X'_{nat})$, it holds that $I(X'_{nat};X'_{pert};F') \le \eta'$. As $\eta' \to 0$, $I(X'_{nat};X'_{pert};F') \to 0$.

According to Theorem 2, we can obtain an approximate of $I(X';F')$ by a linear combination of $I(X'_{nat};F')$ and $I(X'_{pert};F')$ as follows:

$$I(X';F') \approx I(X'_{nat};F') + I(X'_{pert};F'). \tag{13}$$

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 02:28:26 UTC from IEEE Xplore. Restrictions apply.

When we increase $I(X'_{nat}; F')$, $I(X'; F')$ increases accordingly due to the linearity of (13). This allows the model to give different levels of attention to natural and perturbed patterns. We can simply set a trade-off hyperparameter to capture the degree of attention to the two patterns.

### D. Feature-Focusing Adversarial Training

Madry et al. [40] formulate adversarial training as a minimax optimization problem formulated as follows:

$$\min_{\theta} E_{(x,y)\sim D}\left[\max_{\delta\in\Omega}\mathcal{L}(h(x+\delta;\theta),y)\right], \tag{14}$$

where $h$ represents a deep neural network, $\theta$ represents the weight of $h$, $D$ represents a distribution of the clean example $x$ and the ground truth label $y$. $\mathcal{L}(h(x+\delta;\theta),y)$ represents loss function of updating the training model. $\delta$ represents the adversarial perturbation, and $\Omega$ represents a bound, which can be defined as $\Omega = \{\delta : \|\delta\| \leq \epsilon\}$ with the maximum perturbation strength.

As mentioned in Section IV-C, the mutual information $I(X'; F')$ of an adversarial image $X'$ and its feature $F'$ is disentangled into a linear approximation of $I(X'_{nat}; F')$ and $I(X'_{pert}; F')$. This provides the possibility to force the model in adversarial training to pay different levels of attention to the natural and perturbed patterns when ensuring the dependence of the adversarial example on its features. Note that the perturbed pattern is not completely useless and contains some detailed features that help improve clean accuracy. Recall that our goal is to guide the model to learn more core features. To achieve this, we propose two adversarial training strategies: (1) feature concentration, aiming to direct the model's attention toward the core features while minimizing the focus on spurious features, and (2) decision boundary adjustment, aimed at directing the model's decision boundary in alignment with natural patterns while mitigating the risk of overfitting to the decision boundaries of adversarial examples. Based on these two strategies, we sequentially designed Pattern-dependent loss and Natural-margin loss.

*Pattern-dependent loss:* To learn more core features in natural patterns and reduce the impact of spurious features in perturbed patterns, an intuitive way is to weigh the focus on these two patterns by optimizing the following loss function:

$$\mathcal{L} = -I\left(X'_{nat}; E_{\psi}(X')\right) - \beta I\left(X'_{pert}; E_{\psi}(X')\right), \tag{15}$$

where $\beta$ is to weigh the importance of $I((X'_{nat}; E_{\psi}(X'))$ and $I((X'_{pert}; E_{\psi}(X'))$ on the prediction results. However, we found the estimate of $I((X'_{nat}; E_{\psi}(X'))$ is much larger than that of $I((X'_{pert}; E_{\psi}(X'))$, and there is a positive correlation between them. This indicates that adjusting the weights of $I(X'_{pert}; E_{\psi}(X'))$ alone will not bring a significant change in the overall loss, which does not achieve our desired goal. To address this issue, we need to further expand the difference in the dependence of the output on the natural and perturbed patterns. Thus, pattern-dependent loss $\mathcal{L}_{pd}$ is defined, the loss is as follows:

$$\mathcal{L}_{pd} = \mathbb{E}_{\mathbb{P}}\Bigg[-\frac{1}{\tau}I\left(X'_{nat}; E_{\psi}(X')\right) + \log\sum_{i=1}^{N}\exp\left(\frac{I\left(X'^{(i)}_{pert}; E_{\psi}(X')\right)}{\tau}\right)\Bigg], \tag{16}$$

where $\mathbb{E}_{\mathbb{P}}(\cdot)$ denotes the mathematical expectation, $N$ represents the number of randomly selected adversarial examples. Here, $\tau$ is a temperature coefficient. As $\tau$ decreases, the distribution of mutual information corresponding to the natural and perturbed patterns becomes less smooth, the more the model focuses on the natural patterns. $\mathcal{L}_{pd}$ encourages the model to prioritize learning the core features of the natural pattern while disregarding spurious features that are present in the perturbed pattern.

*Natural-margin loss:* The previous works [52], [53], [54] claimed that adversarial examples are a natural consequence of learning decision boundaries. Therefore, decision boundaries play a critical role in clean accuracy and robustness. Since the adversarial examples contain perturbed noise, traditional adversarial training tuning the decision boundaries according to the adversarial examples can improve robustness, but also decrease the clean accuracy due to its over-adjustment [24]. In fact, the natural pattern of adversarial examples can improve the clean accuracy (generalization) of the model's decision boundaries (see Fig. 1). Therefore, we propose a natural-margin loss to make the natural patterns of adversarial examples as far as the decision boundaries.

The margin is defined as a measure of the distance between decision boundaries of different classes. Maximizing the margin aims to enhance the distinction between decision boundaries of different classes as much as possible. We define the margin as follows:

$$mg(X', y) = \left[h\left(X'_{nat}\right)\right]_y - \max_{y'\neq y}\left[h\left(X'_{nat}\right)\right]_{y'}, \tag{17}$$

where $y$ and $y'$ are label indexes, and $[h(X'_{nat})]_y$ means the logit value corresponding to the natural patterns $X'_{nat}$ labeled $y$. The positive value of $mg(X, y)$ in (17) means the distance of the examples correctly classified margin value. However, we find that calculating the mean by (17) cancels out the correctly classified margin value (positive number) and the misclassified margin value (negative number) in a mini-batch, and to solve this problem, the margin loss will be defined as:

$$\mathcal{L}_{mg} = -\mathbb{E}_{\mathbb{P}}\left[\left[h\left(X'_{nat}\right)\right]_y - \max\left[h\left(X'_{nat}\right)\right]\right], \tag{18}$$

where $\mathbb{P}$ is the probability distribution to the natural pattern and $\max[\cdot]$ means the maximum element of the vector. However, from Fig. 5(a), we found that models trained using $\mathcal{L}_{ce} + \mathcal{L}_{mg}$ were more accurate against weak attacks such as PGD and MI-FGSM, and less accurate against strong attacks such as CW and AA. We conjecture this is due to the overfitting of images generated for weak attacks. Therefore, we need to regularize the marginal loss. We further improve using the soft natural-margin

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 02:28:26 UTC from IEEE Xplore. Restrictions apply.

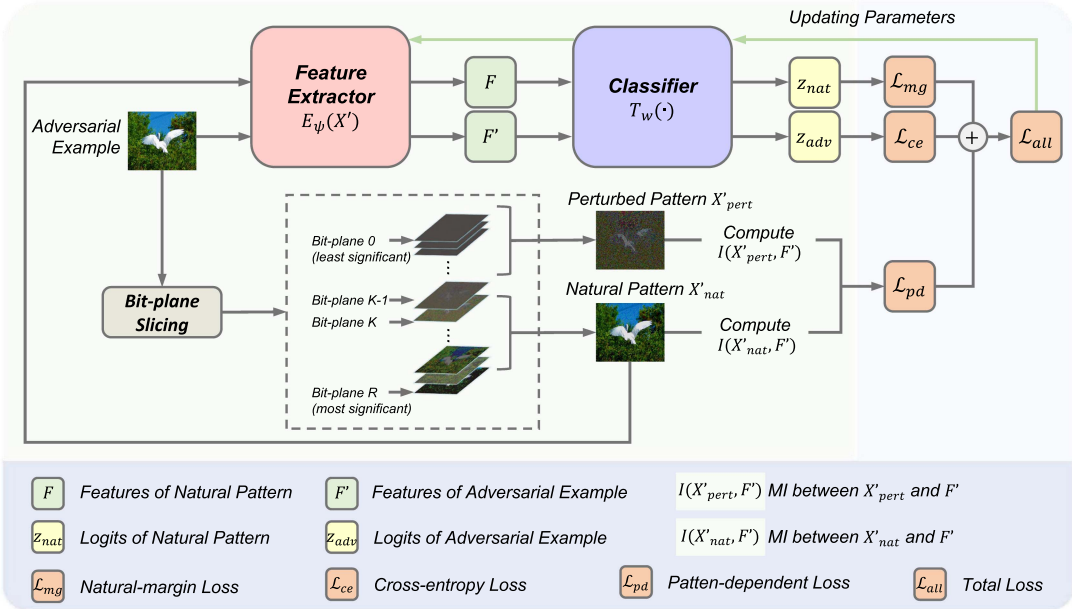



Fig. 4. The overview of our proposed F²AT. Compared to standard adversarial training (SAT) directly conducted on adversarial examples, we first disentangle adversarial examples into natural and perturbed patterns by bit-plane slicing. Through this disentanglement, we further conduct adversarial training on these two patterns. Accordingly, two loss functions, $\mathcal{L}_{pd}$ and $\mathcal{L}_{mg}$, are designed. For $\mathcal{L}_{pd}$, the model is forced to focus on the core features in the natural patterns. For $\mathcal{L}_{mg}$, we force the decision boundary to be far away from natural patterns.

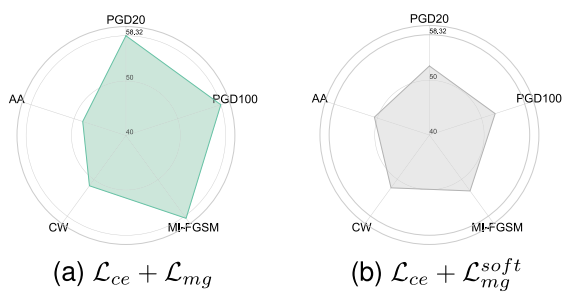



Fig. 5. Adversarial accuracy of models trained against white-box attacks on CIFAR-10 using different losses. The target model is WideResNet-34-10. The models trained using $\mathcal{L}_{ce} + \mathcal{L}_{mg}$ are more accurate against weak attacks such as PGD and MI-FGSM, and less accurate against strong attacks such as CW and AA. The model trained using $\mathcal{L}_{ce} + \mathcal{L}_{mg}^{soft}$ is effective against both weak and strong attacks.

loss of the natural pattern, which is defined as follows:

$$\mathcal{L}_{mg}^{soft} = -\mathbb{E}_{\mathbb{P}}\left[\left[h\left(X'_{nat}\right)\right]_y - \frac{1}{\upsilon}\log\sum_{y'\neq y}\upsilon\cdot\exp\left(h\left(X'_{nat}\right)\right)\right], \tag{19}$$

where $\upsilon$ is the difference smoothing factor. The larger $\upsilon$ is, the less significant the difference between $X'_{nat}$ correctly labeled logit and incorrectly labeled logit will be. In this way, the model is less likely to overfit the natural patterns of adversarial examples, thus increasing the generalization ability of the model.

From Fig. 5(b), the model trained using $\mathcal{L}_{ce} + \mathcal{L}_{mg}^{soft}$ is effective against both weak and strong attacks. Then, we get the total loss of the adversarial training:

$$\mathcal{L}_{all} = \mathcal{L}_{ce} + \alpha\mathcal{L}_{pd} + \gamma\mathcal{L}_{mg}^{soft}, \tag{20}$$

where $\mathcal{L}_{ce}$ is a cross-entropy loss widely used in deep learning. $\alpha$ and $\gamma$ are the trade-off hyperparameters, which play a critical role in balancing the importance of attention to natural and perturbed patterns for a trade-off between clean and robust accuracy. The overview of the algorithm is illustrated in Fig. 4.

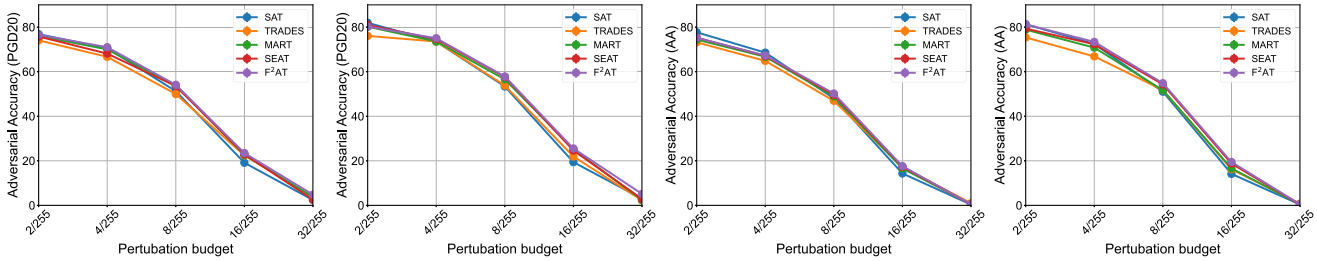
## V. Experiments

### A. Experimental Settings

*Baseline:* Our implementation, built upon the PyTorch framework, underwent rigorous experimentation across various benchmark datasets, including CIFAR-10 [55], CIFAR-100 [55], and Tiny-ImgeNet [56]. The chosen target models are Wide-ResNet-34-10 (WRN-34-10) [57] and ResNet-18 [58]. In the white-box setup, we selected multiple random seeds for our experiments. In the black-box setting, we chose the best-performing model from the white-box setup for testing. To evaluate the black-box setting, the best-performing model from the white-box setup was chosen as the target. The VGG-16 [1] served as the surrogate model while performing transfer attacks. To evaluate the effectiveness of our approach, we compared its performance with several baseline methods, including PGD-based adversarial training (SAT) [9], TRADES [20], MART [21], and SEAT [45].

*Attack setting:* To illustrate the efficacy of our adversarial training, we evaluate adversarial accuracy through both white-box and black-box adversarial attacks. This involves measuring the model's accuracy under various adversarial attack scenarios and the average robustness on the multiple test sets. Specifically, for white-box attacks, we employ four attack methods: PGD [9], MI-FGSM [59], CW [26], and AA [27]. For transfer attacks, we use six attack methods: FGSM [8], PGD [9], TI-FGSM [60], MI-FGSM [59], DI-FGSM [61], and VMI-FGSM [62]. The perturbation budget for white-box and transfer attacks is 8/255. Additionally, PGDT represents a PGD attack with $T$ iteration

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 02:28:26 UTC from IEEE Xplore. Restrictions apply.




(a) ResNet-18 (b) WRN-34-10 (c) ResNet-18 (d) WRN-34-10

Fig. 6. Impact of perturbation budgets $\epsilon$ on adversarial accuracy (AA) and adversarial accuracy (PGD20). It shows that our method can achieve good robustness under different perturbation budgets.

TABLE I
ADVERSARIAL ACCURACY (%) OF ADVERSARIAL TRAINING AGAINST WHITE-BOX ATTACKS

| Dataset | Defenses | None | PGD10 | PGD20 | PGD100 | MI-FGSM | CW | AA |
|---|---|---|---|---|---|---|---|---|
| CIFAR-10 | SAT [9] | **84.88 ± 0.11** | 51.50 ± 0.47 | 51.51 ± 0.35 | 51.49 ± 0.45 | 52.77 ± 0.38 | 50.88 ± 0.39 | 48.38 ± 0.43 |
| | TRADES [20] | 80.80 ± 0.31 | 50.12 ± 0.06 | 50.06 ± 0.11 | 50.04 ± 0.11 | 50.86 ± 0.03 | 48.21 ± 0.03 | 46.94 ± 0.02 |
| | MART [21] | 81.57 ± 0.18 | 53.71 ± 0.73 | 53.73 ± 0.02 | 53.73 ± 0.15 | 54.60 ± 0.10 | 50.46 ± 0.33 | 48.99 ± 0.15 |
| | SEAT [45] | 81.11 ± 0.47 | 53.68 ± 0.13 | 53.60 ± 0.06 | 53.66 ± 0.10 | 54.29 ± 0.17 | 51.40 ± 0.19 | 49.71 ± 0.16 |
| | **F²AT (ours)** | 81.93 ± 0.30 | **54.09 ± 0.13** | **54.06 ± 0.11** | **54.10 ± 0.16** | **54.72 ± 0.06** | **51.99 ± 0.03** | **50.21 ± 0.19** |
| CIFAR-100 | SAT [9] | **58.70 ± 0.12** | 27.97 ± 0.28 | 27.96 ± 0.21 | 27.97 ± 0.24 | 28.67 ± 0.28 | 26.90 ± 0.13 | 24.59 ± 0.06 |
| | TRADES [20] | 54.07 ± 0.23 | 26.56 ± 0.03 | 26.66 ± 0.16 | 26.64 ± 0.12 | 27.01 ± 0.18 | 23.21 ± 0.28 | 22.31 ± 0.08 |
| | MART [21] | 53.64 ± 0.30 | 31.14 ± 0.18 | 31.10 ± 0.16 | 31.10 ± 0.23 | 31.47 ± 0.34 | 28.24 ± 0.25 | 26.36 ± 0.06 |
| | SEAT [45] | 55.26 ± 0.11 | 31.03 ± 0.33 | 31.00 ± 0.22 | 30.97 ± 0.32 | 31.45 ± 0.28 | 28.64 ± 0.21 | 26.62 ± 0.09 |
| | **F²AT (ours)** | 56.24 ± 0.91 | **31.16 ± 0.25** | **31.21 ± 0.23** | **31.16 ± 0.27** | **31.61 ± 0.37** | **28.96 ± 0.40** | **26.80 ± 0.35** |
| Tiny-ImageNet | SAT [9] | 51.30 ± 0.56 | 24.48 ± 0.62 | 24.46 ± 0.55 | 24.50 ± 0.57 | 25.18 ± 0.43 | 21.82 ± 0.34 | 18.72 ± 0.21 |
| | TRADES [20] | 51.04 ± 0.23 | 23.79 ± 0.84 | 23.77 ± 0.79 | 23.78 ± 0.80 | 24.24 ± 0.52 | 19.16 ± 0.07 | 17.46 ± 0.08 |
| | MART [21] | 42.52 ± 0.98 | 25.20 ± 0.39 | 25.18 ± 0.36 | 25.17 ± 0.43 | 25.30 ± 0.58 | 21.40 ± 0.22 | 19.50 ± 0.15 |
| | SEAT [45] | 49.78 ± 0.31 | 25.72 ± 0.49 | 25.34 ± 0.58 | 25.70 ± 0.36 | 25.31 ± 0.33 | 22.94 ± 0.03 | 19.74 ± 0.09 |
| | **F²AT (ours)** | **51.96 ± 0.90** | **25.78 ± 0.38** | **25.66 ± 0.31** | **25.82 ± 0.45** | **26.16 ± 0.54** | **23.04 ± 0.29** | **20.14 ± 0.26** |

The target model is ResNet-18.

TABLE II
ADVERSARIAL ACCURACY (%) OF ADVERSARIAL TRAINING AGAINST WHITE-BOX ATTACKS

| Dataset | Defenses | None | PGD10 | PGD20 | PGD100 | MI-FGSM | CW | AA |
|---|---|---|---|---|---|---|---|---|
| CIFAR-10 | SAT [9] | 86.71 ± 0.75 | 53.39 ± 0.03 | 53.37 ± 0.01 | 53.36 ± 0.02 | 54.63 ± 0.18 | 53.92 ± 0.26 | 51.29 ± 0.35 |
| | TRADES [20] | 84.65 ± 0.14 | 53.98 ± 0.37 | 54.05 ± 0.35 | 54.06 ± 0.45 | 55.15 ± 0.35 | 53.11 ± 0.07 | 51.88 ± 0.16 |
| | MART [21] | 85.66 ± 0.51 | 56.34 ± 0.47 | 57.26 ± 0.95 | 57.31 ± 0.93 | 58.17 ± 0.84 | 55.04 ± 2.47 | 51.75 ± 0.05 |
| | SEAT [45] | 85.53 ± 0.38 | 57.64 ± 0.08 | 57.61 ± 0.07 | 57.72 ± 0.04 | 58.20 ± 0.06 | 56.35 ± 0.22 | 54.50 ± 0.13 |
| | **F²AT (ours)** | **86.88 ± 0.14** | **57.93 ± 0.03** | **57.70 ± 0.21** | **57.74 ± 0.23** | **58.50 ± 0.20** | **56.70 ± 0.09** | **54.86 ± 0.12** |
| CIFAR-100 | SAT [9] | 63.01 ± 0.28 | 29.16 ± 0.45 | 29.34 ± 0.38 | 29.46 ± 0.40 | 30.27 ± 0.32 | 29.55 ± 0.52 | 26.83 ± 0.40 |
| | TRADES [20] | 57.54 ± 0.14 | 30.37 ± 0.06 | 30.32 ± 0.06 | 30.35 ± 0.04 | 30.74 ± 0.06 | 27.83 ± 0.34 | 26.91 ± 0.23 |
| | MART [21] | 58.14 ± 0.13 | 32.55 ± 0.25 | 32.53 ± 0.30 | 32.58 ± 0.31 | 32.97 ± 0.22 | 29.71 ± 0.08 | 27.93 ± 0.07 |
| | SEAT [45] | 60.96 ± 0.78 | 34.73 ± 0.15 | 34.76 ± 0.16 | 34.74 ± 0.16 | 35.22 ± 0.20 | 32.60 ± 0.30 | 30.58 ± 0.21 |
| | **F²AT (ours)** | **63.91 ± 1.18** | **34.88 ± 0.04** | **34.89 ± 0.05** | **34.87 ± 0.06** | **35.30 ± 0.18** | **32.84 ± 0.23** | **30.78 ± 0.24** |
| Tiny-ImageNet | SAT [9] | 54.10 ± 0.41 | 23.38 ± 0.43 | 23.34 ± 0.68 | 23.40 ± 0.59 | 24.08 ± 0.36 | 22.36 ± 0.21 | 19.34 ± 0.47 |
| | TRADES [20] | 54.24 ± 0.38 | 28.44 ± 0.51 | 28.45 ± 0.37 | 28.42 ± 0.37 | 29.18 ± 0.62 | 25.14 ± 0.09 | 22.46 ± 0.20 |
| | MART [21] | 47.38 ± 1.53 | 28.16 ± 0.92 | 28.14 ± 0.82 | 28.20 ± 0.96 | 28.28 ± 0.70 | 25.56 ± 0.11 | 23.00 ± 0.18 |
| | SEAT [45] | 54.24 ± 0.60 | 28.96 ± 0.77 | 28.94 ± 0.71 | 28.92 ± 0.68 | 29.52 ± 0.72 | 26.35 ± 0.23 | 24.76 ± 0.14 |
| | **F²AT (ours)** | **55.20 ± 1.71** | **29.51 ± 0.44** | **29.48 ± 0.46** | **29.50 ± 0.57** | **30.08 ± 0.53** | **27.04 ± 0.28** | **25.16 ± 0.25** |

The target model is WRN-34-10.

steps and a step size of 0.007. For query attacks, we use SimBA [63], SignOPT [64], Square Attack [65], and Hop-skipjump [66] to assess model performance. we set the query limit to 1000 and the perturbation budget to 0.031.

*Defense setting:* We configured the training process with 110 epochs, incorporating a transition epoch set at [100, 105]. This decision followed the configuration in [67]. All images underwent standard data augmentation during training, including random crops with a 4-pixel padding and random horizontal flips [55]. To ensure a fair comparison, we employed SGD for all methods, setting a momentum to 0.9, a weight decay to $2 \times 10^{-4}$, and a batch size to 128. The initial learning rate is set to 0.1, reduced by a factor of 10 at the 100th and 105th epochs. Furthermore, we fine-tuned hyperparameters for the defense methods, establishing $K = 2, N = 1, \gamma = 1, \tau = 0.07$, $\upsilon = 0.995$, and $\alpha = 0.05$.

### B. Performance Under White-Box Attacks

In white-box settings, all attacks have access to the architecture and parameters of the target model. For a comprehensive evaluation, we conducted multiple experiments using different random seeds, and the averaged results are presented in Tables I and II. Furthermore, we investigated the impact of perturbation

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 02:28:26 UTC from IEEE Xplore. Restrictions apply.

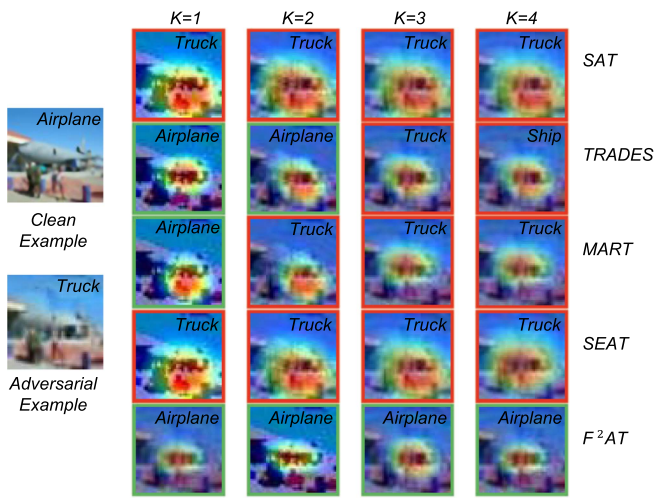



Fig. 7. Class activation maps for adversarial examples. We use PGD20 to attack one of the images in CIFAR-10. The surrogate model is the adversarially trained VGG-16. Regions that appear redder represent areas of the image that have a greater influence on the classification decision for a specific class. Although each baseline misidentifies adversarial examples with the source labeled "airplane" as "truck", a well-trained model can still classify them correctly under different k-corresponding natural patterns ($F^2$AT).

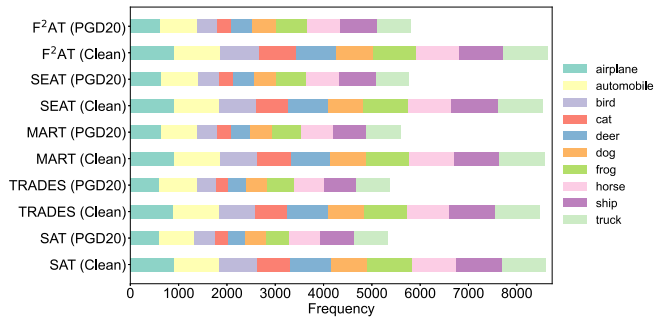



Fig. 8. Distribution of the correct predictions of different adversarial training methods for each class on CIFAR-10 using WRN-34-10. We can infer that our method predicts the frequency of each class more evenly.

budgets under different model structures (see Fig. 6). The results indicate that our method is more effective for small perturbations. Additionally, the prediction frequency of our method is more balanced for each class (see Fig. 8). Although $F^2$AT demonstrates promising performance by leveraging natural patterns in adversarial examples to extract core features, its robustness improvements may be limited when facing attacks like CW and AA, which can generate perturbations violating these underlying assumptions.

To enhance performance, we have employed different training pipelines for our model refinement. Without additional data, we adhere to the training pipeline in DyART [70] and SCORE [69]. For the baseline methods, the epoch has been fixed at 120 while keeping other parameters unchanged. Additionally, we have incorporated an extra dataset in [71] generated by the Diffusion Model [68] to improve the model's performance. Following the training pipeline in [71], we have adjusted the epoch to 400 while retaining the remaining settings unaltered.

TABLE III
ADVERSARIAL ACCURACY (%) OF ADVERSARIAL TRAINING AGAINST WHITE-BOX ATTACKS

| Method | Generated Data | Clean | PGD10 | PGD20 | AA |
|---|---|---|---|---|---|
| SCORE [69] | None | 85.49 | 56.63 | 55.81 | 52.19 |
| DyART [70] | None | 70.21 | 39.87 | 38.84 | 38.23 |
| **$F^2$AT (ours)** | None | **86.88** | **57.83** | **57.62** | **54.59** |
| SCORE [69] | ✓ | 87.19 | 62.34 | 61.60 | 57.21 |
| DyART [70] | ✓ | 79.24 | 47.15 | 46.07 | 45.42 |
| **$F^2$AT (ours)** | ✓ | **87.51** | **63.05** | **62.28** | **59.13** |

The target model is WRN-34-10. The results are obtained by adversarial training WRN34-10 on the CIFAR-10 dataset or CIFAR-10 1M generated by DDPM [68].

The results are presented in Table III. These results highlight the effectiveness of our approach in enhancing clean and adversarial accuracy.

### C. Performance Under Black-Box Attacks

Under a black-box attack, the attacker does not know anything about the internal structure of the attacked model, training parameters, defense methods, etc., except for interacting with the model through the output. The results presented in Tables IV and V demonstrate the superior robustness of our method compared to other baseline techniques. These results highlight the effectiveness of our approach in mitigating adversarial attacks and elevating the resilience of the models in the black-box setting.

### D. Visualization of Feature Focusing

To intuitively reflect the core features of the adversarial image that can be more focused after training the model with our method, we use a visualization way to display them in Fig. 7. Specifically, we generate a series of adversarial images under the black-box setting. We utilized Grad-CAM [72] to obtain the class activation maps for one of them. Fig. 7 demonstrates that the natural patterns we propose contribute to feature focusing, and our method consistently outperforms other approaches in this respect. It emphasizes the effectiveness of our method in capturing and highlighting the image's core features.

### E. Hyper-Parameter Analysis of $K$

Recall that we disentangle an adversarial image into natural and perturbed patterns through bit-plane slicing. The slicing parameter $K$ plays an important role, which determining which bit-planes belong to natural patterns and which ones belong to perturbed patterns. The impact of varying $K$ on the model's accuracy is demonstrated in Table VI. Comparing the scenarios of $K = 2$ and $K = 8$, we observe that incorporating natural patterns can enhance clean accuracy while maintaining model robustness.

### F. Ablation Study

To investigate the role of individual algorithmic components in improving adversarial robustness, we conducted the ablation study in three different cases: $F^2$AT(i) removing natural and perturbed patterns, $F^2$AT(ii) setting the hyperparameter $\gamma$ to

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 02:28:26 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ADVERSARIAL ACCURACY (%) OF DEFENSE METHODS AGAINST BLACK-BOX ATTACKS

| Dataset | Defense | Transfer Attack | | | | | | Query Attack | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | FGSM | PGD20 | TI-FGSM | MI-FGSM | DI-FGSM | VMI-FGSM | SimBA | Signopt | Square | Hopskipjump |
| CIFAR-10 | SAT [9] | 66.55 | 65.16 | **72.60** | 65.21 | 65.85 | 65.12 | 5.08 | 9.72 | 60.13 | 10.73 |
| | TRADES [20] | 62.70 | 61.45 | 67.05 | 61.52 | 61.57 | 61.35 | 1.80 | 4.06 | 56.97 | 7.38 |
| | MART [21] | 63.63 | 62.52 | 68.48 | 62.46 | 62.65 | 62.30 | 9.14 | 9.45 | 60.01 | 11.64 |
| | SEAT [45] | 63.64 | 62.18 | 69.37 | 62.33 | 62.45 | 62.09 | 10.17 | 10.02 | 60.52 | 12.90 |
| | **$F^2$AT(ours)** | **66.74** | **65.33** | 71.91 | **65.22** | **66.13** | **65.49** | **10.75** | **12.79** | **60.88** | **14.44** |
| CIFAR-100 | SAT [9] | 40.93 | 40.53 | 44.79 | 40.37 | 40.84 | 40.17 | 7.29 | 0.73 | 33.87 | 2.14 |
| | TRADES [20] | 38.23 | 37.74 | 40.26 | 37.71 | 37.80 | 37.50 | 3.84 | 0.69 | 29.10 | 1.91 |
| | MART [21] | 38.42 | 37.70 | 41.29 | 37.59 | 37.91 | 37.55 | 10.50 | 0.74 | 33.01 | 2.72 |
| | SEAT [45] | 39.58 | 38.70 | 43.17 | 38.91 | 39.13 | 38.14 | 8.63 | 0.74 | 34.58 | 2.15 |
| | **$F^2$AT(ours)** | **40.99** | **40.67** | **45.45** | **40.92** | **41.02** | **40.83** | **10.66** | **1.01** | **34.70** | **3.67** |
| Tiny-ImageNet | SAT [9] | 38.40 | 38.08 | 41.82 | 37.90 | 38.46 | 37.86 | 17.73 | 0.39 | 32.72 | 1.34 |
| | TRADES [20] | 35.33 | 34.91 | 39.27 | 34.74 | 35.57 | 34.68 | 15.54 | 0.15 | 19.72 | 1.49 |
| | MART [21] | 32.38 | 32.52 | 34.92 | 32.26 | 32.86 | 32.22 | 15.38 | 0.32 | 27.92 | 1.32 |
| | SEAT [45] | 37.96 | 38.06 | 41.42 | 37.82 | 38.22 | 37.60 | 15.22 | 0.24 | 30.98 | 1.62 |
| | **$F^2$AT(ours)** | **38.90** | **39.32** | **42.76** | **38.90** | **38.61** | **38.90** | **18.06** | **0.83** | **32.82** | **1.74** |

The target model is ResNet-18. The best results are indicated in bold.

TABLE V
ADVERSARIAL ACCURACY (%) OF DEFENSE METHODS AGAINST BLACK-BOX ATTACKS

| Dataset | Defense | Transfer Attack | | | | | | Query Attack | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | FGSM | PGD20 | TI-FGSM | MI-FGSM | DI-FGSM | VMI-FGSM | SimBA | Signopt | Square | Hopskipjump |
| CIFAR-10 | SAT [9] | 72.53 | 71.96 | 77.63 | 71.76 | 72.19 | 71.65 | 3.94 | 3.07 | 65.26 | 9.13 |
| | TRADES [20] | 69.52 | 69.16 | 73.58 | 68.94 | 69.22 | 68.82 | 3.79 | 1.41 | 62.26 | 7.66 |
| | MART [21] | 70.35 | 68.89 | 75.18 | 69.55 | 69.98 | 69.42 | 7.63 | 2.59 | 63.39 | 10.59 |
| | SEAT [45] | 69.93 | 69.26 | 76.09 | 69.05 | 69.48 | 68.87 | 6.71 | 0.49 | 65.70 | 9.58 |
| | **$F^2$AT(ours)** | **73.92** | **74.01** | **79.38** | **74.15** | **74.33** | **74.85** | **10.93** | **4.53** | **66.41** | **10.92** |
| CIFAR-100 | SAT [9] | 46.48 | 46.17 | 49.75 | 45.98 | 46.54 | 45.84 | 3.46 | 0.81 | 36.95 | 1.74 |
| | TRADES [20] | 42.81 | 42.88 | 45.59 | 45.52 | 42.93 | 42.50 | 6.87 | 0.77 | 33.83 | 0.80 |
| | MART [21] | 43.57 | 43.13 | 43.36 | 43.09 | 43.24 | 42.99 | 9.99 | 0.85 | 35.31 | 2.55 |
| | SEAT [45] | 44.45 | 44.20 | 48.92 | 43.85 | 44.40 | 43.68 | 8.20 | 0.83 | 39.16 | 1.95 |
| | **$F^2$AT(ours)** | **46.74** | **46.49** | **49.52** | **46.77** | **46.68** | **46.92** | **11.75** | **0.87** | **39.49** | **2.62** |
| Tiny-ImageNet | SAT [9] | 41.44 | 41.86 | 45.16 | 41.46 | 41.88 | 41.34 | 12.92 | 1.37 | 33.32 | 1.76 |
| | TRADES [20] | 41.58 | 42.04 | 45.12 | 41.80 | 42.26 | 41.70 | 13.39 | 1.27 | 34.14 | 1.78 |
| | MART [21] | 37.46 | 37.78 | 39.84 | 37.60 | 37.98 | 37.58 | 10.78 | 1.29 | 32.44 | 1.56 |
| | SEAT [45] | 43.60 | 43.92 | 47.04 | 43.64 | 43.98 | 43.58 | 12.24 | 1.01 | 37.48 | 2.10 |
| | **$F^2$AT(ours)** | **44.02** | **44.54** | **47.50** | **44.18** | **44.92** | **44.14** | **13.74** | **1.46** | **37.98** | **2.21** |

The target model is WRN-34-10. The best results are indicated in bold.

TABLE VI
THE EFFECT OF PARAMETERS ON THE CLEAN AND ADVERSARIAL ACCURACY OF WRN-34-10 TRAINED BY DIFFERENT METHODS

| Parameter | Clean | AA | Parameter | Clean | AA | Parameter | Clean | AA |
|---|---|---|---|---|---|---|---|---|
| $K = 2$ | **86.88** | **54.86** | $\gamma = 0.3$ | 84.63 | 53.05 | $\alpha = 0.05$ | **86.88** | **54.86** |
| $K = 4$ | 85.61 | 53.26 | $\gamma = 0.5$ | 85.79 | 53.53 | $\alpha = 0.1$ | 85.67 | 53.06 |
| $K = 6$ | 85.44 | 53.32 | $\gamma = 0.7$ | 85.38 | 53.43 | $\alpha = 0.2$ | 85.02 | 53.26 |
| $K = 8$ | 85.50 | 53.15 | $\gamma = 1.0$ | **86.88** | **54.86** | $\alpha = 0.5$ | 84.91 | 51.74 |

The CIFAR-10 dataset was utilized for evaluation. The smaller $K$ is, the better trade-off can be achieved.

0, and $F^2$AT(iii) setting the hyperparameter $\alpha$ to 0in (20). We evaluated the performance of these variants using the PGD20 attack on CIFAR-10. As shown in Fig. 9, the results show that each component of our method contributes.

*Impact of natural and perturbed patterns:* Through carefully analyzing the results shown in Fig. 1, we conclude that natural patterns play a crucial role in augmenting robustness. Removing both natural and perturbed patterns leads to a loss of meaning in the loss function $\mathcal{L}_{pd}$. In such instances, we set $\alpha$ to 0 and substitute clean examples for natural patterns. Comparatively, the clean accuracy and adversarial accuracy of the model trained under this condition experience varying degrees of degradation (see Fig. 9). Specifically, the clean accuracy decreases by 1.38%, while the adversarial accuracy against PGD decreases by 1.60 %. This discrepancy in performance is attributed to the presence of adversarial noise and natural noise in adversarial images relative to natural patterns. Such noise exerts an influence on the model's overall performance.

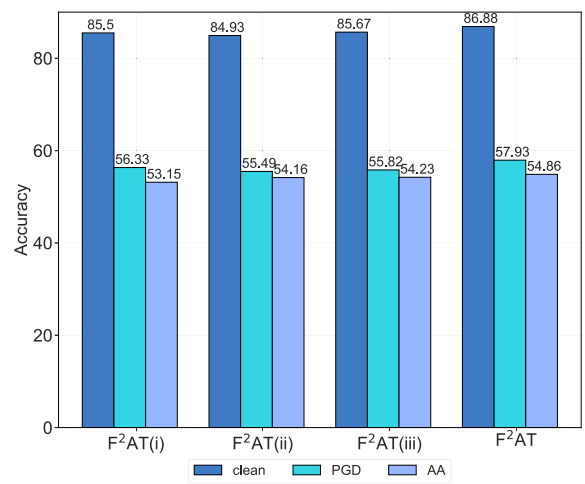



Fig. 9. The ablation study on model WRN-34-10. $F^2$AT(i), (ii), (iii) corresponds to the following three scenarios in turns: (i) removing natural and perturbed patterns, (ii) setting the hyperparameter $\gamma$ to 0in (20), (iii) setting the hyperparameter $\alpha$ to 0in (20).

*Impact of $\mathcal{L}_{mg}$:* The improvement of model clean accuracy and robust accuracy is mainly attributed to $L_{mg}$. When $\alpha = 0$, $\gamma = 0$, it is equivalent to SAT; the larger $\gamma$ is, the more the clean and robust accuracy of the model improves (see Table VI. When we set $\gamma = 0$, the clean accuracy decreased by 1.95%, and the robust accuracy for PGD and AA decreased by 2.44%

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 02:28:26 UTC from IEEE Xplore. Restrictions apply.

and 0.70%, respectively. This shows that focusing on the core features is conducive to improving the robustness and accuracy of the model.

*Impact of $\mathcal{L}_{pd}$:* We further investigate the impact on the experimental results of $\mathcal{L}_{pd}$. Compared to F²AT, our experiments showed that the robust accuracy of F²AT(iii) for PGD and AA decreased by 2.11% and 0.63%, respectively (see Fig. 9). Additionally, the clean accuracy decreased by 1.21%. This result is in line with our intuition of designing the loss: the model's greater attention to the differences between natural and perturbed patterns helps to improve the model's clean and adversarial accuracy. However, when $\alpha$ is excessively large, the model becomes overly focused on features present in the natural pattern but absent in the perturbed pattern. This can result in disregarding useful information within the perturbation pattern, potentially leading to a decrease in accuracy or even failure to converge (see Table VI).

Considering that the examples in the real scenario are not all adversarial, we retrain the model with our method on the clean examples and find that a clean accuracy of 93.83% can still be achieved. This helps to generalize our method to datasets with a mixture of clean and adversarial examples.

## VI. Conclusion

In this paper, we disentangle adversarial examples into natural and perturbed patterns, enabling adversarial training to focus on core features. We propose a novel adversarial training method (F²AT) based on this disentanglement. The experimental results showed that F²AT can significantly improve the adversarial robustness and clean accuracy of the model. In future work, we will further explore the relationship between model robustness and natural perturbations of images. We hope an interpretable and high-accuracy trade-off to be found.

## References


[1] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. Int. Conf. Learn. Representations*, 2015.

[2] M. X. Chen et al., "The best of both worlds: Combining recent advances in neural machine translation," in *Proc. Assoc. Comput. Linguistics*, 2018, pp. 76–86.

[3] Z. Zhang et al., "Deep learning for environmentally robust speech recognition: An overview of recent developments," *Proc. ACM Trans. Intell. Syst. Technol.*, vol. 9, no. 5, pp. 1–28, 2018.

[4] N. Das et al., "Keeping the bad guys out: Protecting and vaccinating deep learning with JPEG compression," 2017, *arXiv: 1705.02900*.

[5] N. Papernot, P. D. McDaniel, X. Wu, S. Jha, and A. Swami, "Distillation as a defense to adversarial perturbations against deep neural networks," in *Proc. IEEE Symp. Secur. Privacy*, 2016, pp. 582–597.

[6] G. S. Dhillon et al., "Stochastic activation pruning for robust adversarial defense," in *Proc. Int. Conf. Learn. Representations*, 2018.

[7] Y. Qian et al., "Hierarchical threshold pruning based on uniform response criterion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 8, pp. 10869–10881, Aug. 2024.

[8] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," in *Proc. Int. Conf. Learn. Representations*, 2015.

[9] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *Proc. Int. Conf. Learn. Representations*, 2018.

[10] C. Zhao et al., "Adversarial training via multi-guidance and historical memory enhancement," *Neurocomputing*, vol. 619, 2025, Art. no. 129124.

[11] X. Liang et al., "Towards desirable decision boundary by moderate-margin adversarial training," *Pattern Recognit. Lett.*, vol. 173, pp. 30–37, 2023.

[12] T. Bai, J. Luo, J. Zhao, B. Wen, and Q. Wang, "Recent advances in adversarial training for adversarial robustness," in *Proc. Int. Joint Conf. Artif. Intell.*, 2021, pp. 4312–4321.

[13] P. Chalasani, J. Chen, A. R. Chowdhury, X. Wu, and S. Jha, "Concise explanations of neural networks using adversarial training," in *Proc. Int. Conf. Mach. Learn.*, PMLR, 2020, pp. 1383–1391.

[14] R. Geirhos, P. Rubisch, C. Michaelis, M. Bethge, F. A. Wichmann, and W. Brendel, "ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness," in *Proc. Int. Conf. Learn. Representations*, 2019.

[15] K. Y. Xiao, L. Engstrom, A. Ilyas, and A. Madry, "Noise or signal: The role of image backgrounds in object recognition," in *Proc. Int. Conf. Learn. Representations*, 2021.

[16] M. Moayeri, K. Banihashem, and S. Feizi, "Explicit tradeoffs between adversarial and natural distributional robustness," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 38761–38774.

[17] A. Ilyas, S. Santurkar, D. Tsipras, L. Engstrom, B. Tran, and A. Madry, "Adversarial examples are not bugs, they are features," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 125–136.

[18] M. T. Ribeiro, S. Singh, and C. Guestrin, ""why should I trust you?": Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discov. Data Mining*, 2016, pp. 1135–1144.

[19] P. Tabacof and E. Valle, "Exploring the space of adversarial images," in *Proc. 2016 Int. Joint Conf. Neural Netw.*, 2016, pp. 426–433.

[20] H. Zhang, Y. Yu, J. Jiao, E. P. Xing, L. E. Ghaoui, and M. I. Jordan, "Theoretically principled trade-off between robustness and accuracy," in *Proc. Int. Conf. Mach. Learn.*, PMLR, 2019, pp. 7472–7482.

[21] Y. Wang, D. Zou, J. Yi, J. Bailey, X. Ma, and Q. Gu, "Improving adversarial robustness requires revisiting misclassified examples," in *Proc. Int. Conf. Learn. Representations*, 2020.

[22] D. Wu, S. Xia, and Y. Wang, "Adversarial weight perturbation helps robust generalization," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 2958–2969.

[23] S. Lee, H. Lee, and S. Yoon, "Adversarial vertex mixup: Toward better adversarially robust generalization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 269–278.

[24] Y. Balaji, T. Goldstein, and J. Hoffman, "Instance adaptive adversarial training: Improved accuracy tradeoffs in neural nets," 2019, *arXiv: 1910.08051*.

[25] C. Szegedy et al., "Intriguing properties of neural networks," in *Proc. Int. Conf. Learn. Representations*, 2014.

[26] N. Carlini and D. A. Wagner, "Towards evaluating the robustness of neural networks," in *Proc. IEEE Symp. Secur. Privacy*, 2017, pp. 39–57.

[27] F. Croce and M. Hein, "Reliable evaluation of adversarial robustness with an ensemble of diverse parameter-free attacks," in *Proc. Int. Conf. Mach. Learn.*, PMLR, 2020, pp. 2206–2216.

[28] P. Chen, H. Zhang, Y. Sharma, J. Yi, and C. Hsieh, "ZOO: Zeroth order optimization based black-box attacks to deep neural networks without training substitute models," in *Proc. 10th ACM Workshop Artif. Intell. Secur.*, 2017, pp. 15–26.

[29] A. Rashid and J. M. Such, "MalProtect: Stateful defense against adversarial query attacks in ML-based malware detection," *IEEE Trans. Inf. Forensics Secur.*, vol. 18, pp. 4361–4376, 2023.

[30] A. Demontis et al., "Why do adversarial attacks transfer? Explaining transferability of evasion and poisoning attacks," in *Proc. 28th USENIX Secur. Symp.*, 2019, pp. 321–338.

[31] Q. Huang, I. Katsman, Z. Gu, H. He, S. J. Belongie, and S. Lim, "Enhancing adversarial example transferability with an intermediate level attack," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 4732–4741.

[32] Y. Qian et al., "Enhancing transferability of adversarial examples through mixed-frequency inputs," *IEEE Trans. Inf. Forensics Secur.*, vol. 19, pp. 7633–7645, 2024.

[33] Y. Gan, C. Wu, D. Ouyang, S. Tang, M. Ye, and T. Xiang, "LESEP: Boosting adversarial transferability via latent encoding and semantic embedding perturbations," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 4, pp. 3368–3382, Apr. 2025.

[34] P. Du, X. Zheng, M. Qi, and H. Ma, "GaitReload: A reloading framework for defending against on-manifold adversarial gait sequences," *IEEE Trans. Inf. Forensics Secur.*, vol. 18, pp. 1380–1392, 2023.

[35] J. Li, T. Xie, L. Chen, F. Xie, X. He, and Z. Zheng, "Adversarial attack on large scale graph," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 1, pp. 82–95, Jan. 2023.


Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 02:28:26 UTC from IEEE Xplore. Restrictions apply.

[36] H. Sun, T. Zhu, Z. Zhang, D. Jin, P. Xiong, and W. Zhou, "Adversarial attacks against deep generative models on data: A survey," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 4, pp. 3367–3388, Apr. 2023.
[37] A. S. Chivukula and W. Liu, "Adversarial deep learning models with multiple adversaries," *IEEE Trans. Knowl. Data Eng.*, vol. 31, no. 6, pp. 1066–1079, Jun. 2019.
[38] F. Liao, M. Liang, Y. Dong, T. Pang, X. Hu, and J. Zhu, "Defense against adversarial attacks using high-level representation guided denoiser," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 1778–1787.
[39] D. Meng and H. Chen, "MagNet: A two-pronged defense against adversarial examples," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, 2017, pp. 135–147.
[40] A. N. Bhagoji, W. He, B. Li, and D. Song, "Practical black-box attacks on deep neural networks using efficient query mechanisms," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 158–174.
[41] H. Kannan, A. Kurakin, and I. J. Goodfellow, "Adversarial logit pairing," 2018, *arXiv: 1803.06373*.
[42] C. Mao, Z. Zhong, J. Yang, C. Vondrick, and B. Ray, "Metric learning for adversarial robustness," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 478–489.
[43] Y. Zhong and W. Deng, "Adversarial learning with margin-based triplet embedding regularization," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 6548–6557.
[44] A. Kurakin, I. J. Goodfellow, and S. Bengio, "Adversarial machine learning at scale," in *Proc. Int. Conf. Learn. Representations*, 2017.
[45] H. Wang and Y. Wang, "Self-ensemble adversarial training for improved robustness," in *Proc. Int. Conf. Learn. Representations*, 2022.
[46] R. McEliece, *The Theory of Information and Coding*. Cambridge, U.K.: Cambridge Univ. Press, 2002.
[47] Y. Sugase, S. Yamane, S. Ueno, and K. Kawano, "Global and fine information coded by single neurons in the temporal visual cortex," *Nature*, vol. 400, no. 6747, pp. 869–873, 1999.
[48] A. P. Sripati and C. R. Olson, "Representing the forest before the trees: A global advantage effect in monkey inferotemporal cortex," *J. Neurosci.*, vol. 29, no. 24, pp. 7788–7796, 2009.
[49] Z. Shan and W. Hai-Tao, "Image retrieval based on bit-plane distribution entropy," in *Proc. 2008 Int. Conf. Comput. Sci. Softw. Eng.*, 2008, pp. 532–535.
[50] D. Zhou et al., "Improving adversarial robustness via mutual information estimation," in *Proc. Int. Conf. Mach. Learn.*, PMLR, 2022, pp. 27338–27352.
[51] L. Paninski, "Estimation of entropy and mutual information," *Neural Comput.*, vol. 15, no. 6, pp. 1191–1253, 2003.
[52] M. Khoury and D. Hadfield-Menell, "On the geometry of adversarial examples," 2018, *arXiv: 1811.00525*.
[53] K. Sun, Z. Zhu, and Z. Lin, "Towards understanding adversarial examples systematically: Exploring data size, task and model factors," 2019, *arXiv: 1902.11019*.
[54] A. Fawzi, S. Moosavi-Dezfooli, and P. Frossard, "Robustness of classifiers: From adversarial to random noise," in *Proc. Adv. Neural Inf. Process. Syst.*, 2016, pp. 1624–1632.
[55] A. Krizhevsky, "Learning multiple layers of features from tiny images," *Handbook Systemic Autoimmune Dis.*, vol. 1, no. 4, 2009.
[56] Z. Kuang, "Tiny-ImageNet visual recognition challenge," 2015. [Online]. Available: https://tiny-imagenet.herokuapp.com/
[57] S. Zagoruyko and N. Komodakis, "Wide residual networks," in *Proc. Brit. Mach. Vis. Conf.*, 2016, pp. 147–148.
[58] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
[59] Y. Dong et al., "Boosting adversarial attacks with momentum," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 9185–9193.
[60] Y. Dong, T. Pang, H. Su, and J. Zhu, "Evading defenses to transferable adversarial examples by translation-invariant attacks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 4312–4321.
[61] C. Xie et al., "Improving transferability of adversarial examples with input diversity," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 2730–2739.
[62] X. Wang and K. He, "Enhancing the transferability of adversarial attacks through variance tuning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 1924–1933.
[63] C. Guo, J. R. Gardner, Y. You, A. G. Wilson, and K. Q. Weinberger, "Simple black-box adversarial attacks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 2484–2493.
[64] M. Cheng, S. Singh, P. H. Chen, P. Chen, S. Liu, and C. Hsieh, "Sign-OPT: A query-efficient hard-label adversarial attack," in *Proc. Int. Conf. Learn. Representations*, 2020.
[65] M. Andriushchenko, F. Croce, N. Flammarion, and M. Hein, "Square attack: A query-efficient black-box adversarial attack via random search," in *Proc. Eur. Conf. Comput. Vis.*, Springer, 2020, pp. 484–501.
[66] J. Chen, M. I. Jordan, and M. J. Wainwright, "HopSkipJumpAttack: A query-efficient decision-based attack," in *Proc. IEEE Symp. Secur. Privacy*, 2020, pp. 1277–1294.
[67] T. Pang, X. Yang, Y. Dong, H. Su, and J. Zhu, "Bag of tricks for adversarial training," in *Proc. Int. Conf. Learn. Representations*, 2021.
[68] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 6840–6851.
[69] T. Pang, M. Lin, X. Yang, J. Zhu, and S. Yan, "Robustness and accuracy could be reconcilable by (proper) definition," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 17258–17277.
[70] Y. Xu, Y. Sun, M. Goldblum, T. Goldstein, and F. Huang, "Exploring and exploiting decision boundary dynamics for adversarial robustness," in *Proc. Int. Conf. Learn. Representations*, 2023.
[71] S. Rebuffi, S. Gowal, D. A. Calian, F. Stimberg, O. Wiles, and T. A. Mann, "Fixing data augmentation to improve adversarial robustness," 2021, *arXiv:2103.01946*.
[72] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-cam: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2017, pp. 618–626.

**Yaguan Qian** received the PhD degree in computer science from Zhejiang University, in 2014. He is currently a full professor with the School of Artificial Intelligence and Information Engineering, Zhejiang University of Science and Technology. He has authored or co-authored more than 40 papers in peer-reviewed international conferences and journals, including ICCV, ECCV, AAAI, *IEEE Transactions on Information Forensics and Security*, *IEEE Transactions on Neural Networks and Learning Systems*, *ACM Transactions on Privacy and Security*, etc. His main research interests include AI security, machine learning, pattern recognition, and machine vision.

**Chenyu Zhao** received the BS degree in data science from the Zhejiang University of Science and Technology, in 2022. She is currently working toward the MS degree with the School of Science, Zhejiang University of Science and Technology. Her research interests include deep learning, machine learning security, and computer vision. She has published one paper as a co-author on ICCV 2023.

**Zhaoquan Gu** received the bachelor's degree in computer science from Tsinghua University, in 2011, and the PhD degree in computer science from Tsinghua University, in 2015. He is currently a professor with the School of Computer Science and Technology, Harbin Institute of Technology (Shenzhen), China. He is also a professor with the Department of New Networks, Peng Cheng Laboratory, Shenzhen. His research interests include cyberspace security, cyber range, Big Data analysis, and artificial intelligence security.

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 02:28:26 UTC from IEEE Xplore. Restrictions apply.

**Bin Wang** received the PhD degree from the China National Digital Switching System Engineering & Technological R&D Center, in 2010. He is professor of the Zhejiang Key Laboratory of Multi-dimensional Perception Technology, Application and Cybersecurity and Zhejiang University. His research interests mainly include Internet of Things security, cryptography, artificial intelligence security, new network security architecture and so on.

**Shouling Ji** received the PhD degree in electrical and computer engineering from the Georgia Institute of Technology, and the PhD degree in computer science from Georgia State University. He is a full professor with the College of Computer Science and Technology, Zhejiang University and a research faculty with the School of Electrical and Computer Engineering, Georgia Institute of Technology (Georgia Tech). His current research interests include data-driven security and privacy, AI security, and Big Data analytics.

**Wei Wang** received the PhD degree from Xi'an Jiaotong University, in 2006. He is a full professor with the School of Computer and Information Technology, Beijing Jiaotong University, China. He was a post-doctoral researcher with the University of Trento, Italy, from 2005 to 2006. He was a post-doctoral researcher with TELECOM Bretagne and with INRIA, France, from 2007 to 2008. He was also a European ERCIM fellow with the Norwegian University of Science and Technology (NTNU), Norway, and with the Interdisciplinary Centre for Security, Reliability and Trust (SnT), the University of Luxembourg, from 2009 to 2011. He has authored or co-authored more than 100 peer-reviewed articles in various journals and international conferences. His recent research interests lie in social networks and data security. He is an Elsevier 'highly cited Chinese Researchers'. He is an editorial board member of *Computers & Security* and a Young AE of *Frontiers of Computer Science*.

**Yanchun Zhang** received the PhD degree in computer science from the University of Queensland, in 1991. He is currently the director of IT Program (Data Science and AI) with the Institute for Sustainable Industries and Liveable Cities, VU Research, Victoria University, Melbourne, VIC, Australia, and coordinates a multidisciplinary e-research program across Victoria University. He is also an International Research Leader in databases, data mining, health informatics, Web information systems, and Web services. He has published more than 300 research articles in international journals and conference proceedings, including *ACM Transactions on Computer and Human Interaction (TOCHI)*, the *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, *VLDB Journal*, SIGMOD, and ICDE conferences, and a dozen of books and journal special issues in the related areas. He is a founding editor and the editor-in-chief of *World Wide Web Journal* (Springer) and *Health Information Science and Systems Journal* (Springer). He is also the chairman of the International Web Information Systems Engineering Society.

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 02:28:26 UTC from IEEE Xplore. Restrictions apply.